Ryan Pratt

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Education

University of Wisconsin–Madison *Expected May 2028*
B.S. in Computer Engineering; Additional Major in Computer Sciences | GPA: 3.73/4.0
Relevant Coursework: Artificial Intelligence, Algorithms, Linear Algebra, Mathematical Statistics; HPC in progress

Skills

Languages: Python, C++, C, Java, SQL, Bash
Frameworks/Libraries: PyTorch, XGBoost, pandas, NumPy, SciPy, FastAPI, JUnit, Mockito
Tools/Platforms: Git, Linux, Docker, Maven, SonarQube, MATLAB, Claude Code
Concepts: Machine Learning, Cross-Validation, Model Evaluation, Feature Engineering, REST APIs, Unit Testing

Experience

Wintrust Financial Corporation – *Full Stack Engineer Intern* *May 2026 – August 2026*
- Increased backend code coverage from 16% to 41% across a roughly 76,000-line Java codebase, developing unit tests with JUnit, Mockito, and AEM Mocks for components, services, and servlets.
- Resolved 400+ SonarQube findings, including critical and major issues, by updating production Java code and deprecated APIs, incorporating peer review feedback, and documenting changes for QA.
- Reduced image payload sizes by approximately 30% by developing a Java OSGi service and integrating WebP delivery into custom Adobe Experience Manager components.

iCode – *Lead Instructor* *June 2025 – August 2025*
- Taught over 200 K–12 students programming and robotics through hands-on projects and guiding debugging.

Technical Experience

Wisconsin Robotics – *Software Team Member* *October 2024 – December 2025*
- Implemented A* path planning in Python for rover navigation, reducing time-to-goal by approximately 15% in simulation and physical testing.
- Integrated navigation software for rover deployment, balancing pathfinding performance with hardware constraints.

Wisconsin AI Safety Initiative – *Technical Safety Scholar* *February 2025 – February 2026*
- Analyzed approximately 60 AI research papers and model releases during a 13-week cohort, evaluating model capabilities, robustness, and failure modes.
- Compared alignment and interpretability approaches across academic and open-source systems, communicating technical tradeoffs and potential mitigation strategies in peer discussions.

Projects

Amino Acid Substitution Prediction – *Machine Learning Marathon 2025* *September 2025 – December 2025*
Tech: *Python, PyTorch, XGBoost, pandas, NumPy, Jupyter Notebook*
- Led a five-person team in a UW–Madison machine learning competition using 175K+ samples, analyzing amino acid substitution data to guide preprocessing and feature selection.
- Reduced cross-validation MSE by 23% by identifying heavy-tailed prediction errors and redesigning training to prioritize high-error cases.
- Compared model variants through cross-validation, selected a more stable final approach, and presented the modeling pipeline, validation strategy, and tradeoffs to 80+ attendees.

Jellyfish – *AI Wiki Search (Badger BuildFest 2026)* *September 2026*
Tech: *Next.js, React, TypeScript, Tailwind CSS, Model Context Protocol (MCP)*
- Won three awards and $2,000 with a four-person team at Badger BuildFest 2026: 2nd place in the Applied AI track, 2nd in the Databricks Real-World Workflows challenge, and a share of 1st in Open Venture.
- Designed the system architecture: a retrieval pipeline that pulls a wide pool of wiki search candidates, re-ranks them with a language model, reuses that ranking to detect pages made stale by an edit, and serves results to Claude through an MCP server.
- Built the product's Next.js/TypeScript web app, showing the ranked sources beside each answer and flagging answers that cite an outdated page, with Yes/No feedback on every answer.